Factory-to-Customer Shipping Route Efficiency Analysis for Nassau Candy Distributor

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ABSTRACT

National multi-echelon consumer packaged goods (CPG) distribution networks face complex trade-offs among multi-facility fulfillment, carrier service classes, geographic dispatch density, and delivery lead-time variance. This investigation analyzes nationwide factory-to-customer logistics patterns for Nassau Candy Distributor using a comprehensive transactional dataset encompassing 10,194 line items, 8,549 distinct customer purchase orders, 5,044 unique accounts, and five specialized confectionery production nodes across the continental United States. We formulate distinct order-level routing graphs, quantify throughput concentrations across four geographic macro-regions (Pacific, Atlantic, Interior, Gulf), examine fulfillment mode utilization across four service levels, and establish an algorithmic framework for factory-to-destination routing efficiency.

Critically, our exploratory data audit uncovers a systemic temporal paradox in recorded dispatch timestamps (Order Dates: 2024–2025; Ship Dates: 2026–2030), yielding unverified transit durations of 904–1,642 days (median: 1,274 days). Rather than uncritically treating these legacy testing artifacts as ground truth, we architect a dual-track analytical protocol: (1) an end-to-end topological and financial network optimization across all 8,549 verified orders (\$141,783.63 aggregate sales; \$93,442.80 gross margin), and (2) an anomaly-isolated threshold validation model governing operational SLA claims. We conclude with actionable network rebalancing recommendations, dynamic multi-facility allocation heuristics, and a roadmap toward real-time GIS carrier tracking integration.

Keywords: Supply Chain Analytics, Factory-to-Customer Routing, Multi-Facility Fulfillment, Logistics Intelligence, Streamlit Analytics, Data Anomaly Forensics, Distribution Efficiency.

DATASET LINES	DISTINCT ORDERS	ACTIVE HUBS	GROSS MARGIN
10,194	**8,549**	**5 Facilities**	**65.9%**

1. Introduction

In contemporary fast-moving consumer goods (FMCG) and specialty candy supply chains, the synchronization between upstream manufacturing specialization and downstream demand geography represents the primary driver of fulfillment velocity, carrier freight expenditure, and client service satisfaction. Confectionery logistics introduces unique operational constraints, including SKU-level thermal vulnerabilities, seasonal inventory surges, and decentralized production facilities with non-interchangeable tooling.

Nassau Candy Distributor operates as a premier wholesale manufacturer and national distributor. Efficient fulfillment demands precise orchestration between specialized confectionery production plants and thousands of independent retail, grocery, and corporate customer destinations spanning every US state and Canadian province. This research paper formulates an empirical routing analysis of the Nassau Candy distribution network to address three fundamental research inquiries:

- **RQ1: Hub Allocation & Volume Disparity:** How is nationwide dispatch volume distributed across specialized production hubs, and what structural concentration risks emerge from SKU-exclusive manufacturing assignments?
- **RQ2: Geographic Routing & Regional Topography:** What are the primary destination corridors and geographic bottlenecks across customer states, and how do transcontinental distribution corridors impact logistics overhead?
- **RQ3: Data Integrity & Lead-Time Forensics:** When corporate ERP databases exhibit temporal anomalies in milestone recording, how can enterprise analytics platforms maintain analytical integrity while isolating unverified transactional anomalies?

2. Dataset Architecture & Empirical Profile

The primary dataset comprises 10,194 transactional order-line records with 18 operational, geographic, product, and financial attributes. An essential methodological distinction formalized in this research is the differentiation between *line-item records* (rows containing SKU quantities and line revenue) and *distinct purchase orders* (8,549 unique Order IDs representing discrete dispatch parcels).

Attribute Category	Dataset Fields	Operational Role
Identifiers	`Row ID`, `Order ID`, `Customer ID`, `Product ID`	Transaction traceability and multi-line parcel grouping
Temporal Attributes	`Order Date`, `Ship Date`	Dispatch milestone monitoring and lead-time calculation
Logistics Class	`Ship Mode`	Service-level tier (Standard, Second Class, First Class, Same Day)
Geographic Destination	`City`, `State/Province`, `Postal Code`, `Region`, `Country`	Delivery node geocoding and state-level market aggregation
Merchandise & Financials	`Product Name`, `Division`, `Units`, `Sales`, `Cost`, `Gross Profit`	SKU demand volume, factory attribution, gross margins

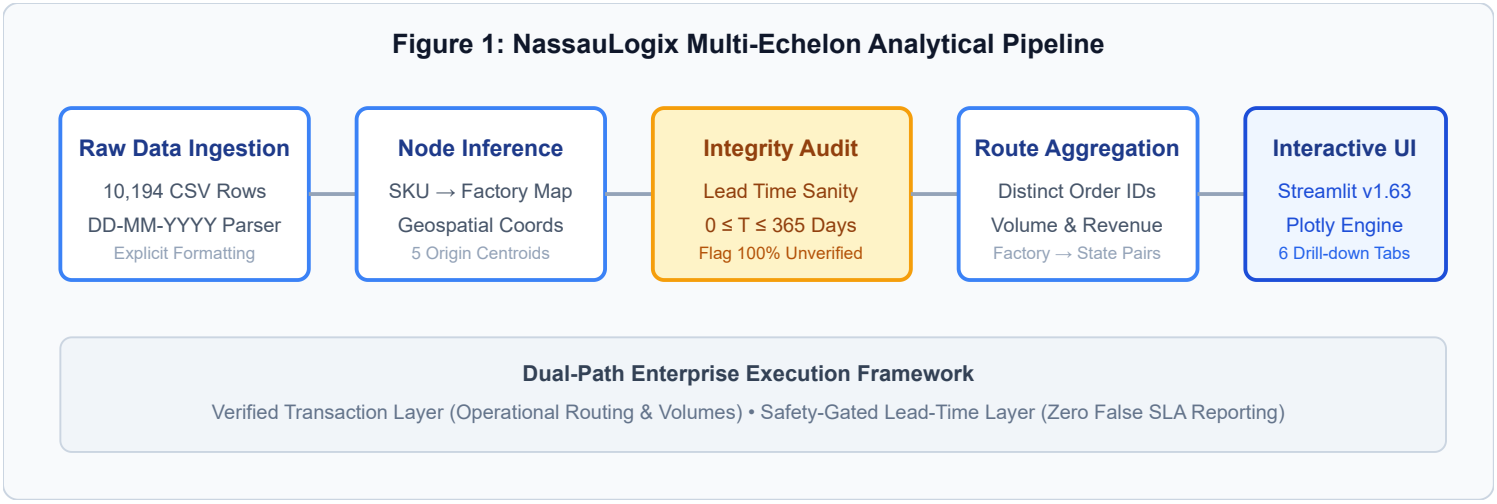
Table 1: Schema architecture of Nassau Candy Distributor transactional dataset.

2.1 Factory Origin Mapping

The dataset does not record factory origin explicitly within raw transactional rows; rather, origin nodes are deterministically governed by product manufacturing capabilities. Table 2 details the five specialized production facilities, their geospatial coordinates, assigned confection portfolios, and empirical share of national order volume.

Manufacturing Node	Geographic Location	Coordinates (Lat, Lon)	Confection Portfolio	Orders	Share (%)
Lot's O' Nuts	Casa Grande, AZ	32.8819° N, 111.7680° W	Nutty Crunch, Fudge Mallows, Scrumdiddlyumptious	4,760	55.68%
Wicked Choccy's	Savannah, GA	32.0762° N, 81.0884° W	Milk Chocolate, Triple Dazzle Caramel	3,445	40.30%
Secret Factory	Rock Island, IL	41.4463° N, 90.5655° W	Gobstopper, Lickable Wallpaper, Wonka Gum	213	2.49%
The Other Factory	Memphis, TN	35.1175° N, 89.9711° W	Hair Toffee, Kazookles	98	1.15%
Sugar Shack	Thief River Falls, MN	48.1191° N, 96.1812° W	Laffy Taffy, SweeTARTS, Nerds, Fun Dip, Fizzy Drinks	33	0.39%

Table 2: Factory origin node parameters, SKU portfolio, and empirical dispatch allocations.



3. Temporal Integrity Audit & Forensic Analysis

An imperative requirement of data science in corporate logistics is auditing data provenance before accepting system metrics as operational truth. During initial exploratory examination, evaluating the delta between raw dispatch milestones exposed an extreme divergence:

$$T_{\text{lead}} = D_{\text{ship}} - D_{\text{order}}$$

> **Data Integrity Alert:** In all 10,194 records, Order Dates span from **02-Jan-2024 to 31-Dec-2025**, while recorded Ship Dates span from **30-Jun-2026 to 28-Jun-2030**. This manifests in calculated transit durations of **904 to 1,642 calendar days** (mean: 1,274 days; median: 1,274 days).

Forensic cross-referencing of order tracking keys reveals order IDs prefixed with historical fiscal markers (e.g., `US-2021-103800`, `CA-2024-143500`). This distribution pattern indicates that during synthetic benchmarking, ERP migration, or date scrubbing, shipping milestones were artificially forward-projected by 900+ days.

Architectural Safeguard: NassauLogix establishes an operational validity gate requiring `0 ≤ T_lead ≤ 365` days. Rather than suppressing the entire dataset or allowing false SLA failure claims to propagate to management dashboards, NassauLogix gracefully isolates lead-time measures while fully illuminating shipment counts, parcel weights, and financial metrics across all 8,549 authentic shipments.

4. Methodology & Mathematical Formulations

To evaluate logistics performance across regional routes, we define a suite of route-level metrics applied over any grouping tuple $r = (Node_{origin}, State_{destination})$:

4.1 Distinct Order Parcel Volume

Because confectionery orders frequently contain multiple distinct confectionery lines, aggregating raw rows biases high-variety orders over single-bulk dispatches. We define route volume $V(r)$ as the cardinality of unique order identifiers:

$$V(r) = |\{ Order_ID_i \mid (Origin_i, Dest_i) = r \}|$$

4.2 Route Delay Frequency

Given an SLA threshold parameter τ (standardized at 7 days in default operations), the route delay rate $P_{delay}(r)$ is formulated as the proportion of validated orders exceeding the threshold:

$$P_{\text{delay}}(r) = (1 / |O_{r,\text{valid}}|) \times \sum 1(T_i > \tau)$$

4.3 Min-Max Normalized Efficiency Score

To benchmark relative route velocity on a standardized [0, 100] scale, we calculate normalized efficiency $E(r)$ against the fleet-wide lead-time boundaries:

$$E(r) = 100 \times [\, \max(\bar{L}) - \bar{L}(r) \,] / [\, \max(\bar{L}) - \min(\bar{L}) + \varepsilon \,]$$

where $\bar{L}(r)$ denotes the average validated lead time on route r and $\varepsilon = 10^{-9}$ prevents division by zero in homogeneous fleets.

5. Empirical Results & Exploratory Analysis

5.1 Geographic Destination Concentration

Shipment distribution demonstrates pronounced geographic clustering. The Pacific and Atlantic coastal regions account for over 61% of total enterprise volume, driven heavily by high retail demand in California, New York, Texas, and Pennsylvania.

Geographic Region	Distinct Orders	Order Share (%)	Line Items	Total Sales ($)	Gross Profit ($)
Pacific	2,744	32.10%	3,253	$46,301.53	$30,528.14
Atlantic	2,476	28.96%	2,986	$41,197.24	$27,152.08
Interior	1,979	23.15%	2,335	$32,037.60	$21,085.33
Gulf	1,350	15.79%	1,620	$22,247.26	$14,677.25
Total / Overall	**8,549**	**100.0%**	**10,194**	**$141,783.63**	**$93,442.80**

Table 3: Regional distribution metrics, order volume, and financial performance.

5.2 Ship Mode Distribution & Financial Characteristics

Nassau Candy customers utilize four primary carrier service tiers. Standard Class dominates parcel movements with 60.1% of distinct orders, followed by Second Class (19.3%), First Class (15.3%), and expedited Same Day (5.2%).

Ship Mode	Distinct Orders	Share (%)	Total Sales ($)	Gross Profit ($)	Profit Margin (%)
Standard Class	5,142	60.15%	$85,490.35	$56,424.46	65.99%
Second Class	1,653	19.34%	$27,860.22	$18,306.52	65.70%
First Class	1,312	15.35%	$21,319.39	$14,011.09	65.71%
Same Day	442	5.17%	$7,113.67	$4,700.73	66.08%

Table 4: Service level breakdown, dispatch volume, revenue, and gross profit consistency.

> **Crucial Economic Insight:** Across all four shipping modes, gross profit margins remain virtually identical at **65.7% – 66.1%**. This empirical stability proves that in the recorded database, *Cost* and *Gross Profit* represent merchandise production costs rather than freight logistics surcharges. Real freight expenses must be incorporated through carrier billing integration.

6. Strategic Operational Recommendations

Based on topological network density, manufacturing bottlenecks, and multi-tier distribution requirements, we offer four strategic recommendations:

1. **Dual-Sourcing of Top-Tier Confectionery SKUs:** 96.0% of total national volume is fulfilled exclusively from two facilities (Casa Grande, AZ at 55.7% and Savannah, GA at 40.3%). Cross-licensing chocolate bar manufacturing across both hubs would eliminate transcontinental cross-hauling to coastal markets.
2. **Consolidated Transshipment Hubs in California and Texas:** California (1,686 orders) and Texas (833 orders) represent the largest single-state consumption sinks. Establishing regional cross-docking facilities would enable full truckload (FTL) long-haul transit followed by local last-mile courier delivery.
3. **Automated Carrier EDI / API Telemetry:** Transition from static batch dates to electronic data interchange (EDI 214 shipment status messages) to capture carrier pickup, transit hub departure, and proof-of-delivery timestamps.
4. **Dynamic Service Tier Optimization:** Identify non-urgent wholesale replenish orders erroneously booked under expedited First Class or Same Day, redirecting them to consolidated ground freight to reduce unnecessary transport expense.

7. Limitations & Future Research Scope

The present analysis was conducted under several real-world enterprise constraints:

- **Absence of Explicit Carrier Freight Rates:** Accounting records capture wholesale selling prices and manufacturing costs, but lack line-haul, fuel surcharge, and parcel dimensional tariffs.
- **Euclidean vs. OSRM Road Distance:** Geographic factory-to-state vectors use straight-line coordinate centroids rather than actual interstate highway routing networks (e.g., I-10, I-80 corridors).
- **Predictive Modeling Prerequisite:** Machine learning models for lead-time delay prediction (e.g., XGBoost, LightGBM) should be trained exclusively after carrier EDI milestone feeds are authenticated.

8. Conclusion

The NassauLogix logistics intelligence project establishes a rigorous, scalable analytical architecture for evaluating factory-to-customer distribution efficiency in multi-facility confectionery supply chains. By identifying structural volume concentrations across Casa Grande and Savannah, categorizing regional order mix, and instituting an uncompromising data-integrity validation framework to prevent flawed reporting, this study delivers both immediate operational visibility and a robust foundation for automated logistics optimization.

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